Sami Mostfa

sami.mostfa@etud.polymtl.ca | linkedin.com/in/sami-mostfa | github.com/tarteeg

Profile

Professional Master's student in **Computer Engineering** at Polytechnique Montréal, I am looking for an internship in Montréal for the **Summer 2027** term (May 3 – August 31, 2027), primarily in **Data Engineering**, **MLOps** or **ML Engineering**, while remaining open to other opportunities.

My dual background in software engineering (**ENSEEIHT**) and AI/ML (**Polytechnique Montréal**) gives me a solid foundation in both model design/training and infrastructure deployment. During my internship at *Axians HPC*, I contributed to the design of a **Kubernetes** architecture for integrating an **LLM** inference server. I now want to apply this dual skill set to an end-to-end project, from design to production, as an **ML Engineer** or **MLOps Engineer**.

Education

Polytechnique Montréal 08/2025 – 08/2027
Professional Master's in Computer Engineering – Montréal, Canada

- Current cumulative **GPA**: **3.83/4.0**
- Coursework: Artificial Intelligence, Machine Learning, Continuous Optimization, Data Mining, Data Visualization, Bayesian Statistics, Natural Language Processing, Intelligent DevOps for Large-Scale Software Systems, Cybersecurity

ENSEEIHT 09/2023 – 06/2025
Software Engineering – Toulouse, France

- Coursework: Software and Systems Engineering, Concurrent and Communicating Systems, Databases, Graph Theory, Automata Theory, Computability & Complexity, Computer Architecture, Object Technology, Networks, Formal Methods, Language Semantics & Translation

Preparatory Classes (CPGE) — Lycée Claude Fauriel 09/2021 – 07/2023
MPI / MP2I (intensive Mathematics, Physics and Computer Science) – Saint-Étienne, France*

Experience

HPC Engineering Intern 06/2025 – 08/2025
Axians HPC (High Performance Computing) *Labège, France*

- **Solution put into production by Axians in 2026**
- Designed an architecture integrating **Kubernetes** with the **Open OnDemand** (OOD) HPC portal, working around the native OOD adapter's limitations (no native support for *Deployments*)
- Developed a **Bash** *kubectl* wrapper that intercepts the adapter's calls and redirects *get/delete* commands to the actual resources, making job status display on the web interface reliable
- Designed **Kyverno** policies for automatic cleanup of orphaned resources, quota management and access hardening (NFS mounts, GPU) on the cluster
- Deployed and configured an **LLM inference server (vLLM)** on the Kubernetes cluster
- Wrote a technical architecture document formalizing the OOD workflows (**Apache**, **NGINX**, **Passenger** and the Kubernetes adapter) for the client and the team
- Hard skills: **Linux systems**, **Kubernetes**, **Bash**

Private Tutor in Mathematics & Computer Science 09/2020 – 06/2025
Superprof – Self-employed *Toulouse, France (Hybrid)*

- Tutored about **15 high-school students** in mathematics and computer science alongside my studies, for a total of about **80 hours**
- Soft skills: **Communication**, **adaptability** to each student, **autonomy** in running an activity over several years

Lifeguard 07/2024 – 09/2024
Escap'ad Aquatic Center *Saint-Laurent-de-Chamousset, France*

- Active pool supervision to ensure swimmer safety and emergency response in case of accidents
- Soft skills: **Responsibility**, **decision-making** in critical situations

Projects

Graph Clustering by Modularity Maximization | *Julia* 07/2026 – Present

- Implemented a scientific paper by Professor Daniel Aloise, instructor of Data Mining (INF8111) at Polytechnique Montréal
- Modularity optimization with several algorithms: **LPA**, **LPAm**, **LPAm+** (classical greedy and multi-step strategies) on 9 benchmark graphs from the DIMACS Implementation Challenge (from 34 to over 31,000 vertices)
- **Reviewed 5 scientific papers** to ground the implementation and validate the results

- Real-world applications: affinity group detection in **social networks**, **financial fraud detection**, understanding protein interactions in **biology**
- Hard skills: **Data Science**, **Graph Theory**, **Applied Mathematics**

Air Quality Forecasting in Montréal for 2025 | *Julia* 01/2026 – 04/2026

- **Team of 5**; graded **95/100** in the course Probabilistic and Statistical Methods for AI (MTH3302), Winter 2026
- Forecast fine particulate matter across **7 Montréal stations** from **17 years** of historical data (about **33,000 samples**)
- Pipeline: variable-specific missing-data handling (temporal interpolation, median imputation, removal), then **feature engineering** of over **20 variables** (lags, rolling means, weather interactions, seasonal and wildfire indicators)
- Compared 4 alternative approaches: time weighting, per-station models, regime separation, VIF/AIC selection
- Selected the best model by **RMSE** minimization and submitted it on **Kaggle**
- Hard skills: **Time Series**, **Linear Regression**, **Feature Engineering**

Basketball Detection and Localization with Deep Learning | *Python* 03/2025 – 05/2025

- **Team of 4**, Deep Learning course (N9EN19C) at ENSEEIHT
- Designed a **multi-output CNN** (4 convolutional blocks, 32 to 256 filters) combining presence **classification** and bounding-box **regression** to locate the ball
- Implemented **from scratch** the **IoU** metric (Intersection over Union), vectorized with TensorFlow tensors, along with Precision/Recall/F1 scores at a 40% overlap threshold
- Multi-task training over **25 epochs** (combined BinaryCrossentropy + MSE loss, Adam optimizer), with image normalization (64x64 px) and bounding-box coordinate standardization
- Real-world applications: **automated sports video analysis**, **real-time object tracking**
- Hard skills: **Deep Learning**, **Computer Vision (CNN)**, **TensorFlow/Keras**

Technical Skills

Languages: Python, Julia, C, Java, TypeScript, JavaScript, OCaml, Ada, Bash, HTML5, CSS3, LaTeX
Data Science: NumPy, Pandas, Scikit-Learn, Seaborn, XGBoost, DataFrames.jl, Distributions.jl, Statistics.jl
Data Visualization: D3.js, Matplotlib, Gadfly.jl, OpenGL
Machine Learning, AI & Deep Learning: PyTorch, TensorFlow, Keras, GLM.jl
Environments: VS Code, Vim, GCP, Google Colab, Eclipse, Jupyter Notebook
DevOps & Infrastructure: Kubernetes, Docker, NGINX, Apache, Kyverno, Ruby
Agentic AI: OpenRouter, MCP (Model Context Protocol), Claude Code, Prompt Engineering, Custom skills & tools development
Data Formats: SQL, CSV, JSON, Parquet, XML, YAML, XLSX, Markdown, TeX
Languages (spoken): French (fluent), English (B2 – Cambridge Linguaskill), Spanish (A2), Arabic (basic)

Interests

Music: 14 years of guitar and advanced music theory — youtube.com/@sami.musiic
First aid & lifesaving: 400 cumulative training hours — French National Lifeguard Certificate (BNSSA, 8 months of training) and First Aid Team Responder Level 1 (PSE1, 40 hours)
Guitar lessons: 10 hours taught, alongside math and computer science tutoring
Sports: Swimming & weightlifting